

# Richie Tran

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## EDUCATION

### Carnegie Mellon University

*B.S. Computer Science and Machine Learning*

Pittsburgh, PA

May 2027

### Relevant Coursework

Data Structures and Algorithms, Computer Systems, AI Representation and Problem Solving, Machine Learning I, Discrete Mathematics, Functional Programming, Ideas in Theoretical Computer Science

## EXPERIENCE

### Software Engineer Intern | *Opportunity Knocks*

October 2025 - Present

- Developed a React Native mobile app with Supabase to bring the existing web application to iOS and Android, targeting 5,000+ users at launch.
- Improved application stability by resolving production bugs, reducing reports, and shipped new features to enhance user experience.

### Research Assistant | *Professor Reid Simmons, CMU*

January 2025 - Present

- Developing Victor the GameBot in Python, a multimodal Scrabble-playing agent integrating computer vision, speech recognition, and audio processing to enable real-time human-robot gameplay and interaction.
- Building a computer vision pipeline to detect and interpret board state, and implementing speech and audio processing modules for natural language trash-talk during gameplay.

### Software Developer | *Scotty Labs, CMU*

August 2024 – Present

- Built an AI-powered presentation tool in React and Node.js, enabling 100+ users to receive real-time feedback on delivery.
- Improved campus accessibility for 1,000+ students by contributing to CMUMaps, CMUResearch, and CMUEats.

### Driverless Control Engineer | *Carnegie Mellon Racing*

August 2024 – Present

- Reduced path-planning latency by optimizing algorithms in C, integrating sensor fusion for autonomous navigation across multiple track configurations.
- Contributed to team's 13th place finish out of 100 teams at Formula FSAE Electric.

### Lead Programmer | *FRC, Clarkstown Cyborgs*

September 2020 – August 2024

- Engineered autonomous path-planning and swerve drive systems, contributing to regional finalist placement at FIRST Robotics Competition.
- Accelerated team development by mentoring 40+ members on software and electrical systems.

## PROJECTS

### MyEyes | *Python, Swift, Flask, CoreML, YOLOv8*

- Developed an audio-based iOS app enabling visually impaired users to navigate safely through real-time hazard detection.
- Achieved 90% detection accuracy by deploying a fine-tuned YOLOv8 model via CoreML with a custom urgency-weighted loss function.
- Integrated LiDAR and camera-based obstacle identification with Siri commands for accessibility.

### AI Logo Generator | *Node.js, Express, Claude API, JavaScript, HTML, CSS*

- Built a full-stack web app that processes user meeting transcripts and generates custom SVG logos and design rationales, leveraging Claude AI to streamline brand identity creation.
- Developed RESTful API endpoints with Express.js to integrate Claude Sonnet 4.5 and Opus 4.5 models for real-time brand analysis and logo generation.
- Implemented drag-and-drop file handling, enabling seamless user workflow.

## SKILLS

**Programming Languages:** C, C++, Python, Swift, Java, JavaScript, HTML, CSS, TypeScript, Bash

**Tools:** Node.js, Express, React, Claude AI, Firebase, Supabase, React Native, Linux, Git, NumPy